

Dao Zhou

Department of Psychology and Behavioral Sciences, Zhejiang University

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EDUCATION

Undergraduate: Zhejiang University (QS global rank 47th), Hangzhou, China 09/2023 – 06/2027

- Overall GPA: 4.17/4.3

Major: *Bachelor of Science in Psychology (Honor's Program)*

Minor: *Mathematics and Applied Mathematics*

Exchange: University of California, Berkeley, Berkeley, USA

Summer 2025

RESEARCH INTERESTS

Language and thought; cognitive neuroscience of language; computational models of language processing; human-LLMs comparison in language cognition; philosophy of language

RESEARCH EXPERIENCE

Project: An MEG Investigation of Perceptual and Semantic Constraints on Language Processing 07/2026 – Present

Leader & Core Collaborator | Advisor: Prof. [Liina Pyllkkänen](#) | [Neurolinguistics Lab](#) @NYU

Investigating perceptual and semantic constraints underlying limits of human sentence comprehension.

- Designing a rapid parallel visual presentation MEG experiment manipulating *visual chunk length* and *information density*.
- Developing stimulus conditions to dissociate neural signatures of *discourse coherence* and *syntactic structure*.
- Localizing structure-sensitive neural responses to determine whether comprehension limits arise from input length or information density.

Project: Understanding Heterogeneity in Neural Activity of Sentence Processing Using Word Embeddings 09/2024 – Present

Leader & Core Collaborator | Advisor: Prof. [Xiang-Zhen Kong](#) | Cognomics Lab @ZJU

Studying the functional heterogeneity within the human language network during sentence processing.

- Collected task-fMRI responses to 1,000 diverse sentences with 10 human-rated psycholinguistic properties.
- Preprocessed fMRI data using CONN and estimated single-trial beta values with GLMsingle.
- Built ROI-wise ridge decoding models across 384 brain regions to predict psycholinguistic features from neural activity.
- Extracted and normalized regression weights across SENSEAAS-defined ROIs to characterize region-specific linguistic profiles.

Project: Optimal Combination of Sample Size and Observation Duration for Behavioral Rate Estimation 12/2025 – Present

Core Collaborator | Advisor: Dr. [Danyang Han](#), Prof. [Karen Adolph](#) | Play and Learning Lab @ZJU

Studying the best behavioral sampling strategy under fixed experimental time.

- Collected infant behavioral data to inform simulations of within- and between-participant variability.
- Simulated sampling strategies across sample sizes and observation durations, identifying optimal designs by minimizing RMSE.
- Extended G-theory models to incorporate preparation time as experimental overhead, optimizing sampling efficiency.

Project: The Mental Number Line is Logarithmic Rather than Linear in the Human Intraparietal Sulcus 04/2026 – 07/2026

Independent Class Project

Studying the linearity of number representation in the human brain.

- Constructed subject-specific Intraparietal Pulcus (IPS) ROIs and neural RDMs.
- Performed ROI-based RSA to test linear versus logarithmic number-line models.
- Integrated Principal Coordinates Analysis (PCoA) with Procrustes alignment to quantify cross-subject stability of neural representational geometry.

Project: Source Languages Systematically Shape the Lexico-Grammatical Profiles of Translated English 04/2026 – 07/2026

Independent Class Project

Studying what impact source languages exert on their translated version.

- Extracted 67 Biber features from 16 translated English varieties and quantified source-English typological distances using the World Atlas of Language Structures (WALS).
- Tested whether typological distance predicts systematic shifts in translated English lexico-grammatical profiles.
- Identified two mechanisms: direct structural mapping for close languages and cognate camouflage for distant languages.

MANUSCRIPTS

■ **Zhou, D.**, Tao X., & Kong, X. (in progress). *Characterizing functional heterogeneity of the human language network during sentence processing*.

■ Han, D., **Zhou, D.**, Herzberg, O., Ossmy, O., Tamis-LeMonda, C., & Adolph, K. E. (in progress). *Estimating the Rate of Behavior: Optimal Combination of Sample Size and Observation Duration*.

CONFERENCE PRESENTATIONS

■ **Zhou, D.**, Tao X., Duan Y., Zhang F., Guo, D., & Kong, X. (2026). *Embedding-based Prediction of Brain Responses Reveals Emotional Effects in Sentence Processing* [Poster Presentation]. **Organization for Human Brain Mapping Annual Meeting**, Bordeaux, France.

■ **Zhou, D.** (2026). *‘Can’ Psychology Be a Science? Reflections on Boundaries and Methodology*. **The 2050 Conference**, Yunqi Town, Hangzhou, China.

■ **Zhou, D.**, Zhang F., Guo, D., & Kong, X. (2025). *Understanding Heterogeneity in Neural Activity of Sentence Processing Using Word Embeddings* [Oral Presentation]. **Undergraduate Research Symposium**, Zhejiang University, Hangzhou, China.

■ **Zhou, D.**, Guo, D., & Kong, X. (2025). *CSCSV: Chinese Sentence Corpus with Semantic Variety* [Poster Presentation]. **International Workshop on Cross-linguistic Databases and Norms**, East China Normal University & The Hong Kong Polytechnic University, Shanghai, China.

PATENT

■ Kong, X., **Zhou, D.**, Li, N., & Pu, Y. (2025). *A Method for Selecting Language Function Rehabilitation Corpus Based on Semantic Embedding and Neural Encoding*. Chinese Patent Application No.CN 202511371878.0, filed September 24, 2025.

FUNDINGS & HONORS

● Undergraduate Future Academic Rising Star Program, Zhejiang University (¥20,000)	2026
● National Innovation Project, Chu Kochen Honors College Research Training Program, Zhejiang University (¥15,000)	2025
● Second Prize Scholarship of Zhejiang University	2025
● Third Prize Scholarship of Zhejiang University	2024
● Dean’s List	2024-2026

SKILLS

■ **Experimental Techniques:** MRI, MEG, EEG, eye-tracking, VR, behavioral experiments

■ **Programming:** Python, MATLAB, R

■ **Software Techniques:** SPM, FreeSurfer, MNE-python, PyTorch, PsychoPy, Praat, PCIBex, Linux

■ **Language:** English (TOEFL: 107; GRE: Verbal 164, Quant 170), Mandarin (native)

■ **Hobbies:** Western philosophy (especially Wittgenstein), film, literature, music, Chinese calligraphy